



Hazards Planning and Risk Management

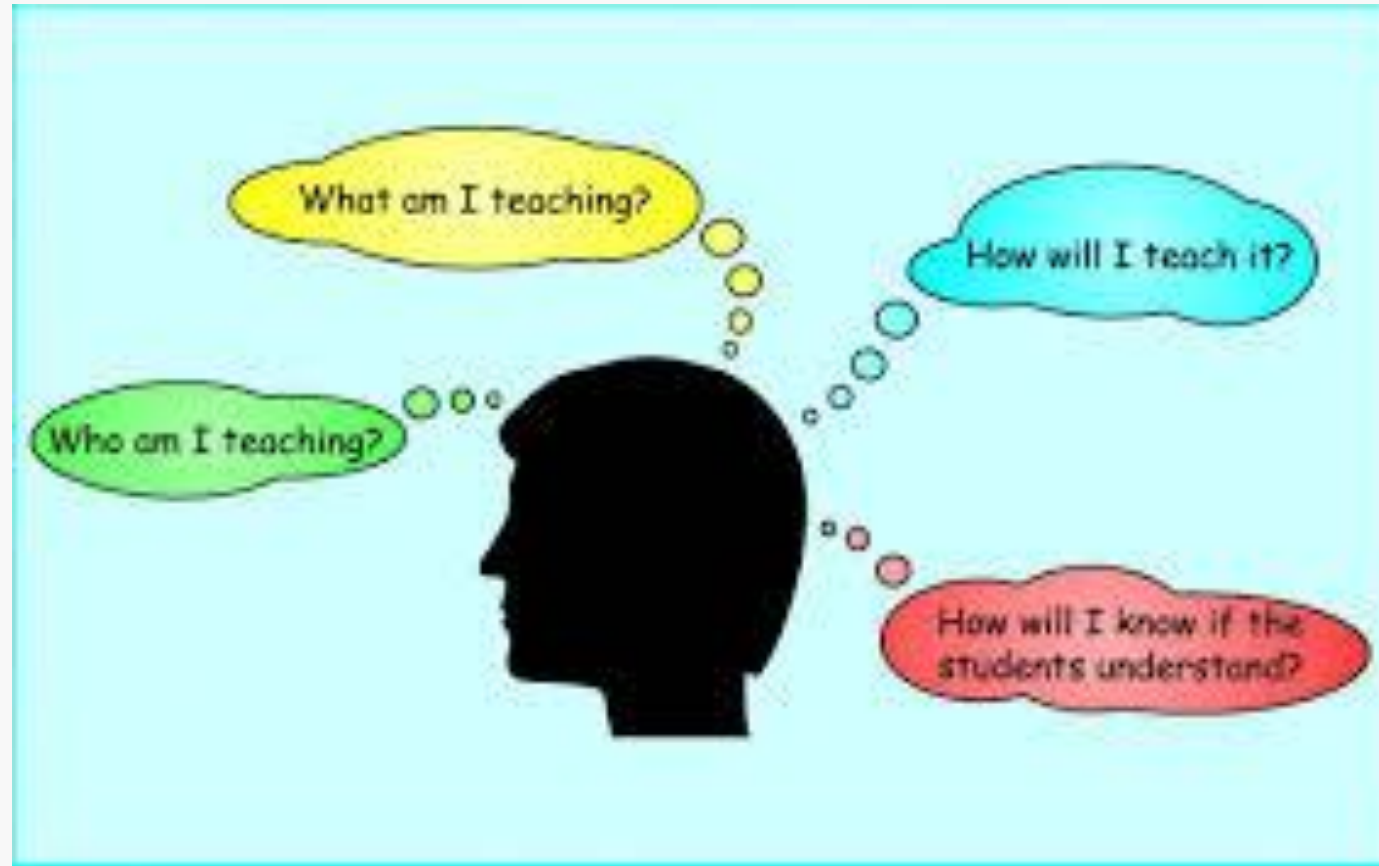
LECTURES 1 & 2

INTRODUCTION TO COURSE

Fall 2026

US – Pakistan Center for Advanced Studies in Water

Objective of today's class?



To introduce the course outline, reading material, purpose, and aim of this course, and other course-related activities

Learning outcomes of the course

After completion of this course, you will be able to

1. LEARN the concepts of hazards and disasters (H&D) management



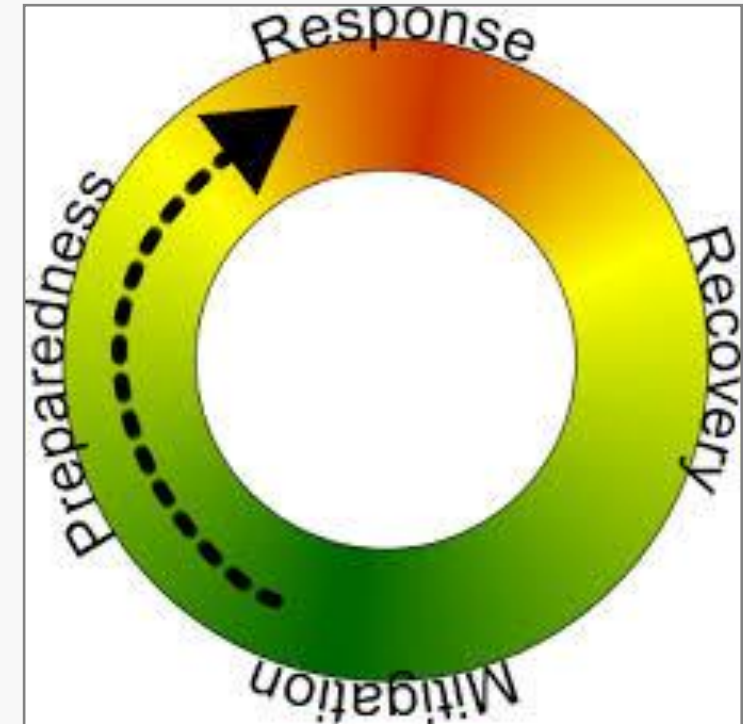
Learning outcomes of the course

2. UNDERSTAND the causes and characteristics of natural hazards and related disasters.



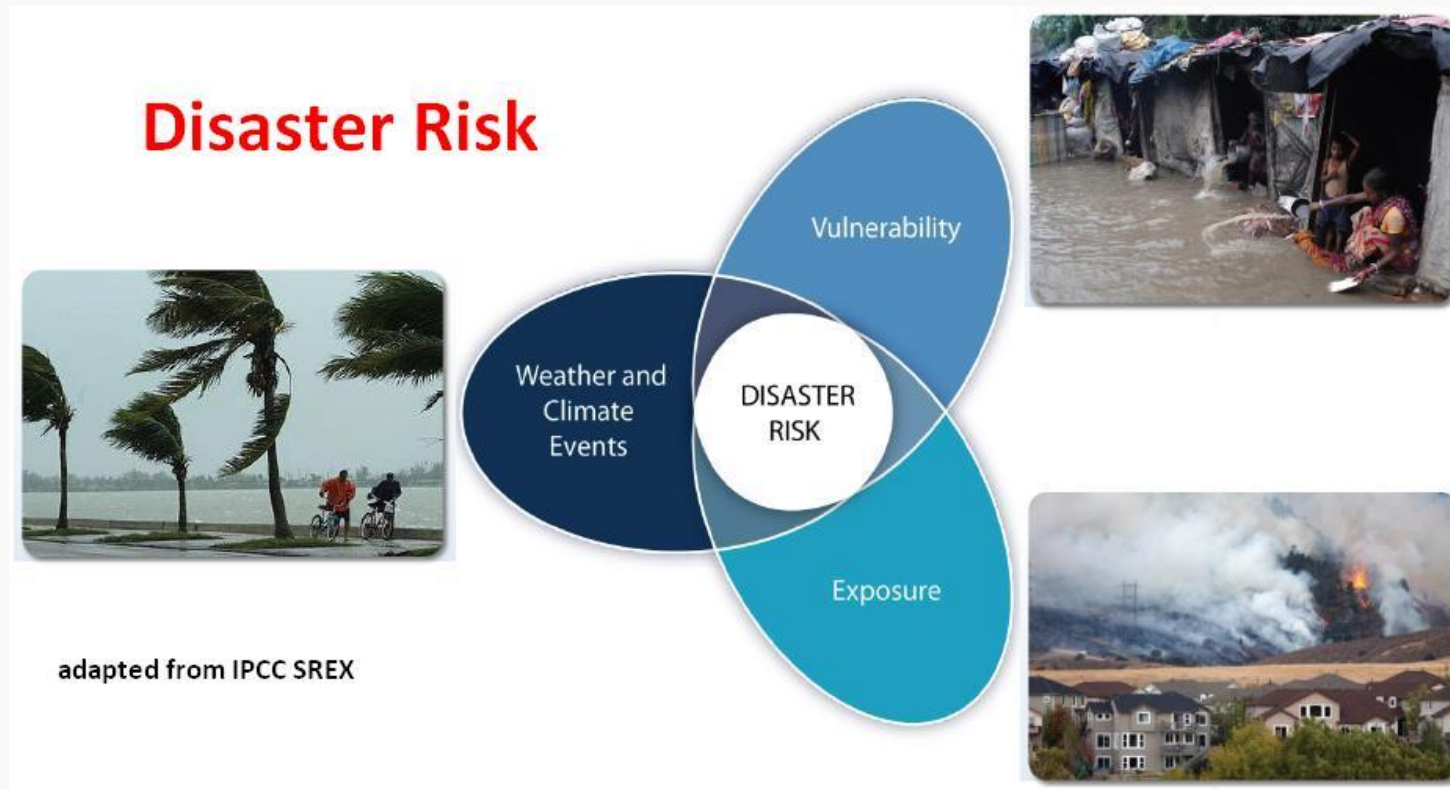
Learning outcomes of the course

3. DISTINGUISH- between different phases of H&D management



Learning outcomes of the course

4. **APPLY** hazard management knowledge to recognize, assess, and deal with disaster risks.



Learning outcomes of the course

5. **LEARN** meaningful strategies for disaster prevention, mitigation, response, and recovery.



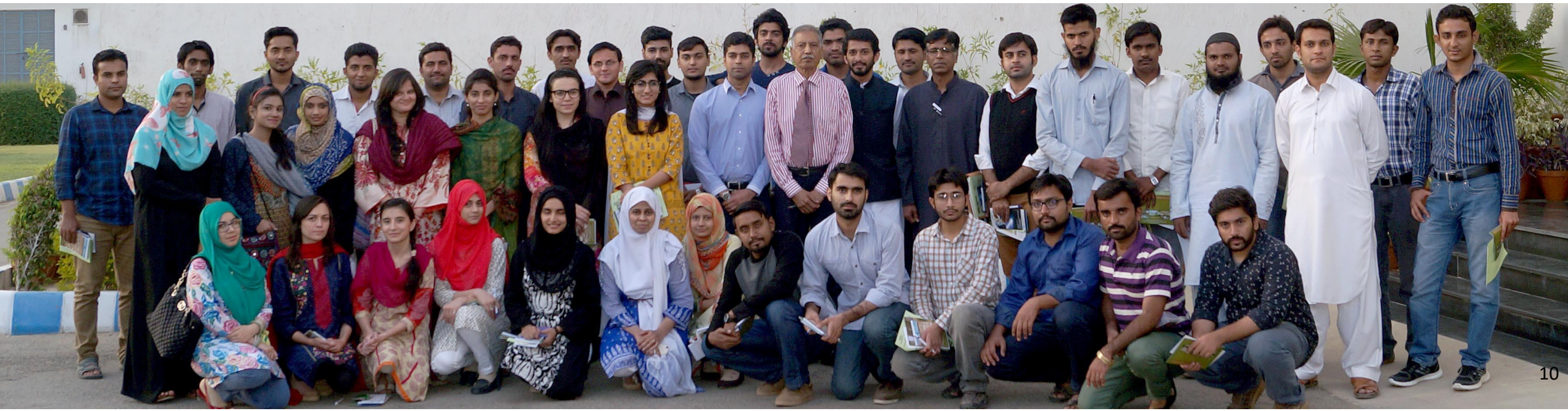
Learning outcomes of the course

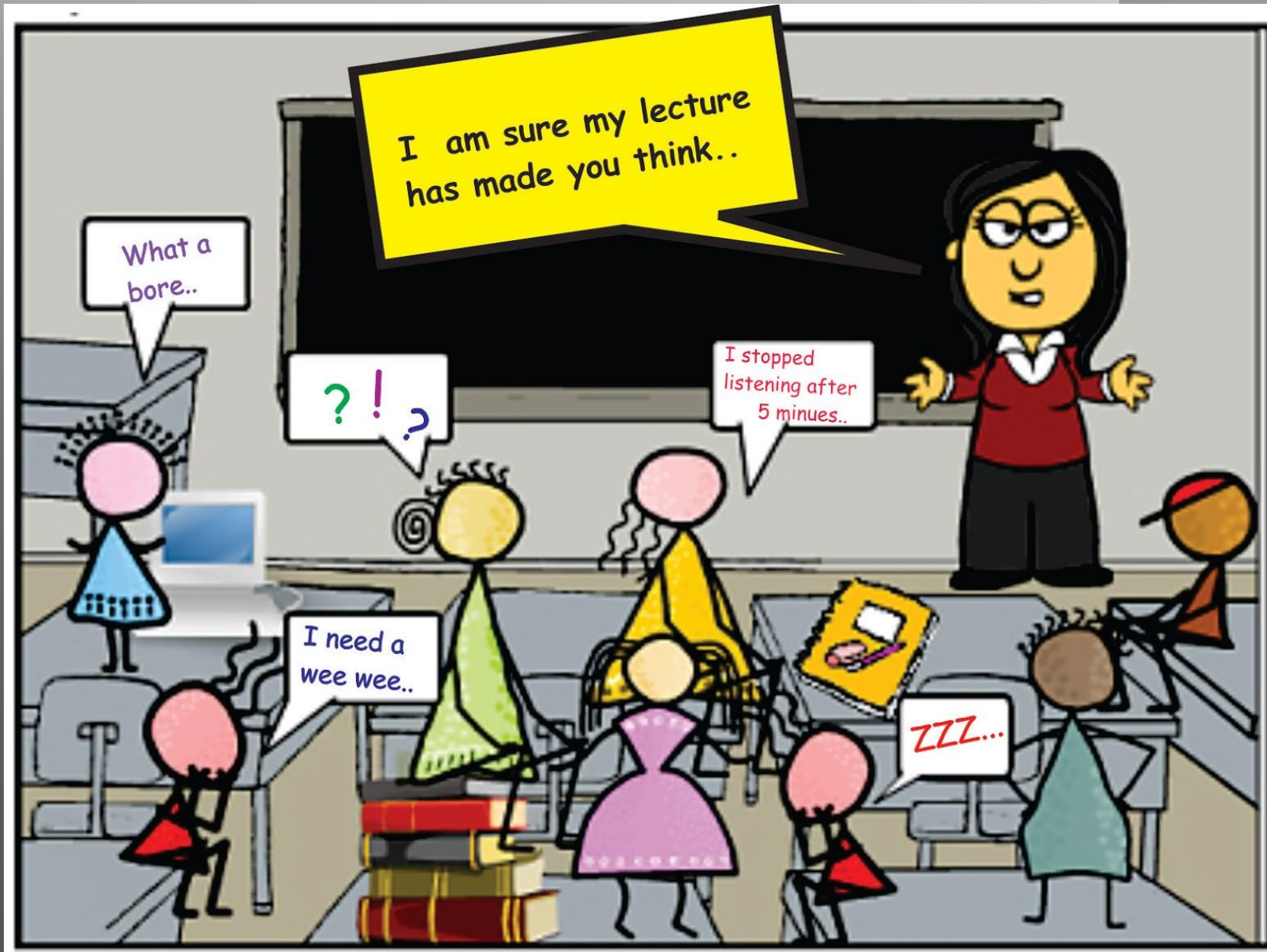
6. USE GIS and satellite data in different phases of hazard management.



What I expect from you?

- Ask questions
- Read as much as possible
- Do all your HWs
- Participate in class activities
- Proactively involve in project activities
- **Be on time and punctual in the class**





How can we make
class environment
conducive of
learning and
friendly?

Class Activity



What do students want from their teachers?



What do you think about RESPECT???



Let's Discuss!

What are your expectations from this course?

Have you experienced or studied any disasters before?

How do you think technology can improve disaster response?

A requirement of the course!!
Group Projects and Presentations

SOME STUDENTS' PROJECTS

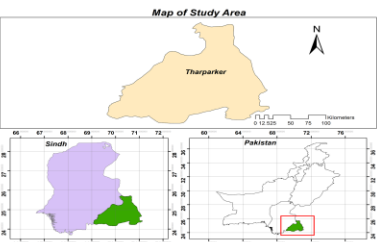
Introduction

Drought is a slowly evolving natural disaster, which grows slowly and can occur in almost any climatic condition. The time of its occurrence is closely linked to the amount of rainfall. However, each drought incident is unique in its cause and effect (Li and Xiao 1992, Wilhite 1993). Frequency, time duration, and extent are three main features by which a drought can be defined (Wilhite and Glantz 1985, American Meteorological Society 1997). Pakistan also has drought-prone areas including the Tharparkar district. Tharparkar is the largest district of the Sindh province by land area. Due to insufficient and erratic rain patterns, the frequency of drought is high in the desert of Tharparkar. The latest drought is ongoing and has taken a heavy toll on people living in the area. For effective hazard planning, drought monitoring in the Tharparkar district is needed. Emerging geospatial techniques can be helpful in this regard.

Objectives

- To analyze the vegetation cover using Normalized Difference Vegetation Index (NDVI) during 2010 - 2015.
- To monitor soil moisture using Temperature Vegetation Dryness Index (TVDI) for the same years.

Study Area



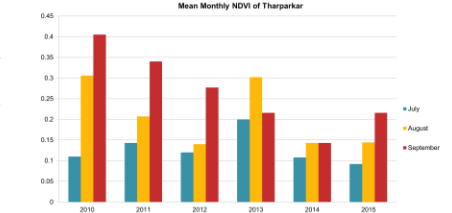
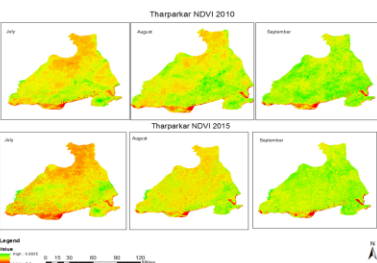
Methodological Framework



Results

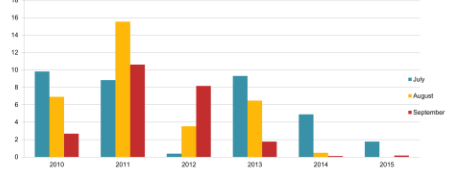
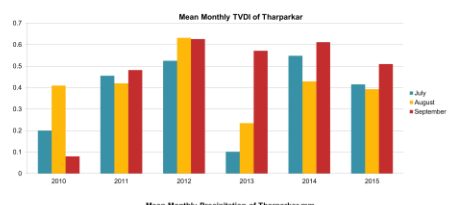
Monitoring NDVI

NDVI is a good indicator of green biomass, leaf area index, and patterns of production. NDVI is the most commonly used vegetation index; its value varies from +1 to -1. Following figure presents the NDVI map of the Tharparkar district.



Monitoring Soil Moisture

Soil moisture is an essential factor to be considered for drought monitoring. It can also provide information on crop water stress and yield estimation. The combination of vegetation index and land surface temperature can provide useful insight of soil moisture status at regional level. The potential of Temperature-Vegetation-Dryness Index from Moderate Resolution Imaging Spectroradiometer (MODIS) data in assessing soil moisture was investigated in this research. Figure below shows the mean monthly TVDI of the study area.



Conclusions

NDVI and rainfall have a close relationship with each other. Due to the changing climatic pattern, the mean annual rainfall in the Tharparkar district has decreased causing severe droughts in the region. For monitoring the droughts, TVDI can be a useful technique.

The months of July in all study years have the least NDVI and TVDI, therefore, July can be considered as the most drought-prone month following low precipitations in Tharparkar.

References

- Chen, Jian, et al. "Estimating soil moisture using Temperature-Vegetation Dryness Index (TVDI) in the Huang-huai-hai (HHH) plain." *International Journal of Remote Sensing* 22.4 (2001): 1165-1177.
- Wan, Z., Wang, P., & Li, X. (2004). Using MODIS land surface temperature and normalized difference vegetation index products for monitoring drought in the southern Great Plains, USA. *International journal of remote sensing*, 25(1), 63-72.
- Kise, J., Jin, J., & Carbone, G. J. (2010). Monitoring agricultural drought for arid and humid regions using multi-sensor remote sensing data. *Remote Sensing of Environment*, 114(12), 2875-2887.

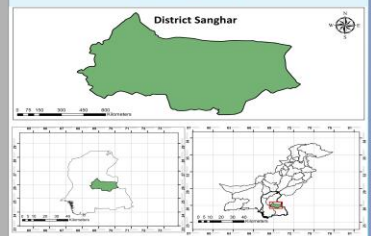
Introduction

Sanghar, an eastern district of Sindh, has a barren land. The population is mainly dependent on agriculture. Cotton, sugarcane, and wheat are the chief cash crops of the district. In 2010-2011, the area was subjected to flash floods due to heavy rainfall. Floods may adversely affect the economy by destroying standing crops, but on the other hand, they increase the fertility of the agriculture land. This study aims to quantify the effect of floods on the vegetation cover of the barren land. After analyzing through geospatial tools, it was found that the vegetation cover has increased to a significant extent.

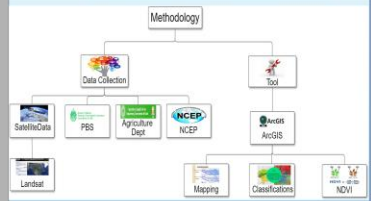
Objectives

- To map floods extent of 2010 and 2011 in the Sanghar district situated on the left bank of the River Indus.
- To calculate change in the vegetation cover before and after the floods.

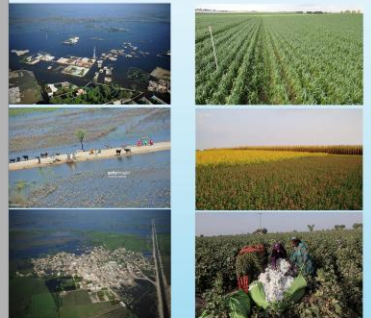
Study Area



Methodology Framework



During flood

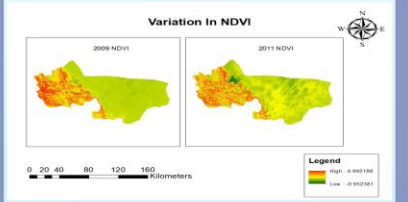


Source: PHOTOCREDIT: Reuters, The Express Tribune, Farooq MANSUR, PHOTOCREDIT: AP, Chaudhry

http://www.amsu.edu.pk/Agriculture/Drought/Water-2012-2018/Sanghar-2012-2018/Wheat.html
http://www.amsu.edu.pk/Agriculture/Drought/Water-2012-2018/Wheat.html
http://www.amsu.edu.pk/Agriculture/Drought/Water-2012-2018/Wheat.html

Subject:- Hazard Planning and Risk Management
Student Mentor:- Syed Talal Naseer
Course Instructor:- Dr. Arjumand Zaidi

Results



Limitations

The image of 2012 is not used in our research because of the excessive cloud cover.

Conclusions

- According to Pakistan Met Department's report of 2012, massive floods have inundated major areas of vegetation cover where crops were ready to harvest, and the flash flood of 2011 damaged all the crops.
- Although crops were severely affected during the flood, soon after the flood, an increase in the vegetative cover presented a positive impact of the disaster.
- The study results also validated the report and showed a significant increase in the vegetation cover after the flooding years.

References

- http://www.amsu.edu.pk/Agriculture/Drought/Water-2012-2018/Sanghar-2012-2018/Wheat.html
- http://www.amsu.edu.pk/Agriculture/Drought/Water-2012-2018/Wheat.html
- http://www.amsu.edu.pk/Agriculture/Drought/Water-2012-2018/Wheat.html

INTRODUCTION

This study is an effort to evaluate the impact of water distribution through a link canal on agriculture in the areas near one of the abandoned rivers in eastern Punjab, Pakistan. Sutlej, the longest river of the Indus basin, originates from west of lake Rakashastal in Tibet and runs through the Himachal Pradesh state, India. After receiving many tributaries, the river finally enters Pakistan at about 15km west of the Kasur District (Iqbal, 2018). After the partition of the subcontinent in 1947, the distribution of Indus water among the two newly born countries became a challenge given rise to many cross-border conflicts. Finally, the Indus Water Treaty(1960) was signed between the two countries and the eastern rivers – Sutlej, Ravi, and Bias were surrendered by Pakistan to India. The security of the fertile lands of eastern Punjab, including the districts of Pakpattan, Arifwala, Burewala, and Bahawalnagar was affected due to the drying of the rivers and their corresponding canals and distributaries (Iqbal, 2018). It was decided in Article IV(1) of the Treaty that as an alternative to these rivers, Pakistan was to carve five new link canals to help ensure water provision to areas falling near them (The IWT, 1960). The Balloki-Sulemanki (B-S) link-canal was built for this purpose in 1951 and enlarged in 1960 under the treaty that connects the Ravi River to the Sutlej River. The main goal of this study is to ascertain if the link canal is playing its role in maintaining the fertility of eastern Punjab of Pakistan or not? Temporal changes in the vegetation cover in the surrounding areas along a selected river reach are determined after the functioning of the link-canal. The study also identified the impact of river flows through the link canal on the vegetation cover.

HYPOTHESIS

Abandoned Sutlej River is drying while entering in Pakistan and the area associated to river is being converted in desert due to vanishing of vegetation cover.

METHODOLOGICAL FRAMEWORK

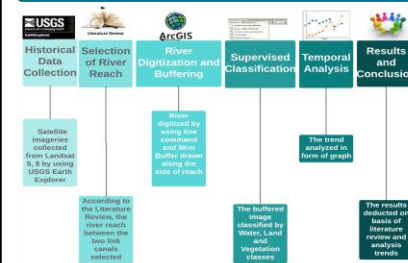


Fig 1. Methodology

RESULTS

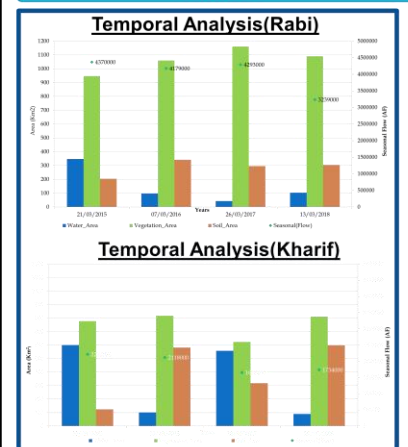
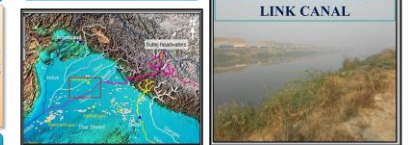


Fig 2. Temporal and Seasonal Analysis

Acknowledgment: This is to acknowledge the efforts of Dr. Arjumand Z. Zaidi and Mr. Salman Sarwar who helped us in this project.

STUDY AREA MAP



TEMPORAL ANALYSIS

Historical satellite images were acquired for temporal-trend analysis of vegetation cover present along the selected river segment. Maximum likelihood supervised classification was performed on the multi-year images of Landsat 5 and 8 to delineate Landcover classes of water, vegetation, and soil (Table 1).

Table 1.

Date	Imageries	Water (Km ²)	Vegetation (Km ²)	Soil (Km ²)	Total (Km ²)	Seasonal Flow
23/02/1994	Available	134.31	836.79	528.45		
19/09/1994	Available	113.13	630.24	755.92		
30/03/1995	Available	41.99	910.79	546.55		
06/09/1995	Available	122.99	738.71	637.63		
18/03/2014	Available	1024.12	189.54	285.79		
26/09/2014	Available	61.30	888.96	549.36		

Note: Imageries of the period 1960-1984 are not available.

DISCUSSIONS AND CONCLUSIONS

The results proved the study hypothesis to be false by indicating the presence of vegetation cover in the study area. The study further suggested that Pakistan had saved the suspected area from drying through management by constructing B-S link canal.

No continuous trend was analyzed in the vegetation area during the last four years, which creates a gap in our study due to the unavailability of historical water flows. This necessitates the need for further studies after acquiring the missing data.

REFERENCE

- Spatio-Temporal and Physiographical Study of the Abandoned Sutlej River: A Case of Jhangi Wala, Bahawalpur, Pakistan, Muhammad Javed Iqbal, 2018.
- The Indus Water Treaty, 1960

INTRODUCTION

Phytoplankton are the base of marine food chain. They help to sustain life in the ocean through photosynthesis. They play an important role in using carbon dioxide from the ocean during photosynthesis and producing oxygen as a by-product [1]. *Noctiluca scintillans* generally called “sea sparkle” is increasing in number and forcing out planktonic species that mostly support the Arabian sea food web. It was unheard of in the Arabian sea 20 years ago. The sparkle created by *Noctiluca scintillans* can be seen by people as colored glow or bloom in the water, which shows up when the water is disturbed [2].

OBJECTIVES

- ☐ To identify the algal blooms in Arabian sea
- ☐ To temporally compare the presence of different color patterns
- ☐ To discuss the possible reasons causing these blooms and its impacts on fish and human beings

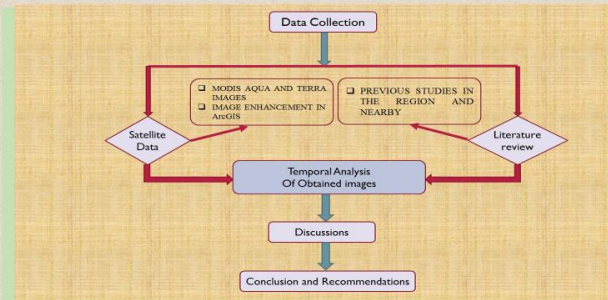
STUDY AREA



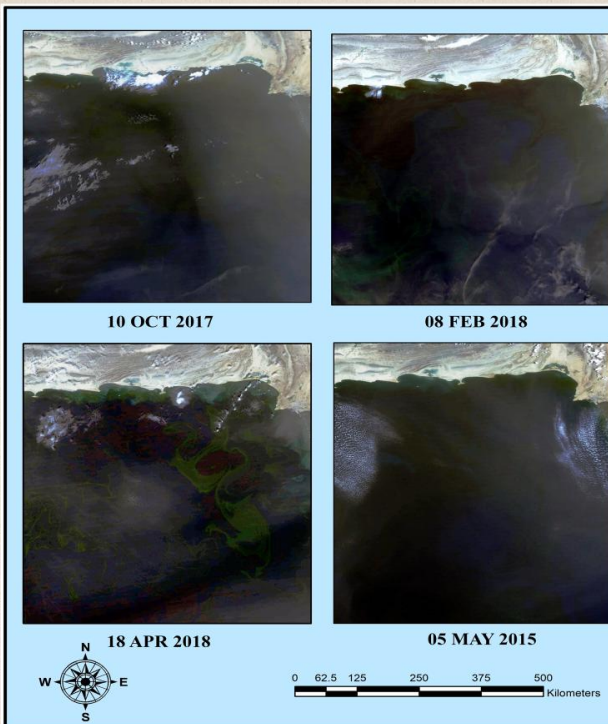
IMAGE ENHANCEMENT

ArcGIS is used to visualize the blooms. Histogram specification stretch is applied to visually enhance the satellite images. A stretch increases the visual contrast of the raster display.

METHODOLOGY



SATELLITE DATA



DISCUSSIONS

Noctiluca scintillans, a typical algal species, turns the water green. One explanation behind the algal growth is climate change. Other reasons are the flood waters that bring minerals from land into the ocean; providing nutrients for phytoplankton to grow. In this study different pictures were obtained from Moderate Resolution Imaging Spectrometer (MODIS) on NASA's Aqua and Terra satellites. These algal blooms look attractive but are a major reason to worry. The thick algae can suffocate fish, cause illness and may prove fatal for humans as it travels up the food chain causing exchange of poison. Some more damages of these blooms include the reduction in visibility causing difficulties for divers.[3] .

RESULTS

- ☐ The algal blooms were clearly visible in the month of February and was seen as green color wavy patterns.
- ☐ The temporal comparison between images of different months of the year showed algae in the month of October that reaches to its peak in February and disappears by start of May.
- ☐ The extent of bloom could not be ascertained in August and September images due to cloud cover , but its extent in October image implied its presence in the earlier two months which are the peak fish catch months [4].

CONCLUSION AND RECOMMENDATIONS

Pollution and other human activities in the coastal zone have increased the quantity of toxic algae. There is a need to understand the correlation between these extensive algal growth and environmental degrading behavior by men, to manage and develop proper mitigation strategies. Thus, control on untreated sewage discharges into the ocean can be stated as one of the reasons behind abnormal growth of harmful algae. Future studies can further this work by considering other possible factors

REFERENCES

1. Khan, Imran Ahmed, et al. "Assessing spatial and temporal variability in phytoplankton concentration through chlorophyll a satellite data: A case study of northern Arabian Sea." *Pak. J. Bot* 47.2 (2015): 797-805.
2. Lamont-Doherty Earth Observatory annual report 2017
3. <https://mic.com/articles/171204/arabian-sea-algae-is-beautiful-but-it-spells-disaster-for-our-planet#.AlivKsGmT>
4. Mr. Ejaz Fisherman. Telephonic conversation.

ACKNOWLEDGEMENT

This is to acknowledge the efforts of Dr.Arjumand Zaidi, Mr. Babar and Mr.Faris Bhatti who helped us in our project.



Is modification in the precipitation pattern affecting the Hanna Lake (Quetta) area?

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Department of Integrated Water Resources Management

US-Pakistan Center for Advanced Studies in Water, MUET

SUBJECT: - Hazard Planning & Risk Management SUPERVISOR: - Dr. Arjumand Zaidi



INTRODUCTION

- In the past 17 years, the Hanna Lake, located in Balochistan, has dried several times. Hanna used to be the most visited and easily accessible lake of Balochistan that has now changed into a barren land [1].
- This situation is very alarming since lakes can affect many essential components of the hydrologic cycle and influence many aspects of ecology and human welfare [2].
- Last year this lake has no water and it was presumed that it was due to less than an average annual rainfall in the region.
- Better understanding and detection of temporal changes in the lake area is important for the assessment and conservation of water resources.
- Solution are needed for sustainability of the Lake such as recharging ground water by building dams and controlling extraction of underground water as heavy rainfall will only fill the lake on a temporary basis [3].

OBJECTIVES

- 1) To determine the relationship between annual precipitation amount and the area of Hanna Lake, Quetta, Balochistan
- 2) To investigate temporal variation in lake area and identify the causes of its drying.

STUDY AREA

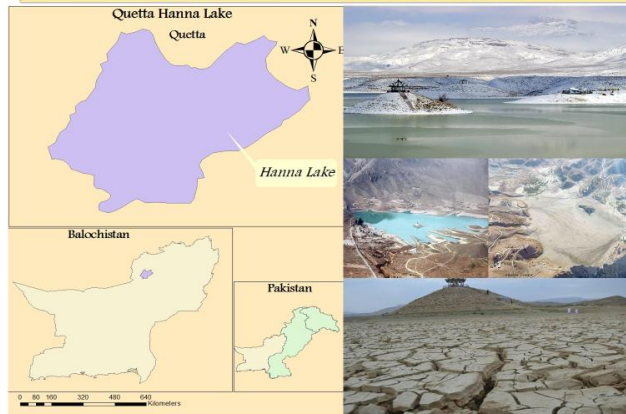


Figure 1. Map of study area



Figure 2. Temporal changes of lake with time

METHODOLOGY

- The meteorological data from 1984 – 2017 were collected from National Oceanic and Atmospheric Administration (NOAA) and normalized to plot annual precipitation.
- Further, historical Lake sizes for the years of 1984 – 2017 were calculated for December months using Google Earth archived images. The temporal Lake areas were digitized, measured, mapped, and plotted against annual precipitation.



RESULTS AND DISCUSSIONS

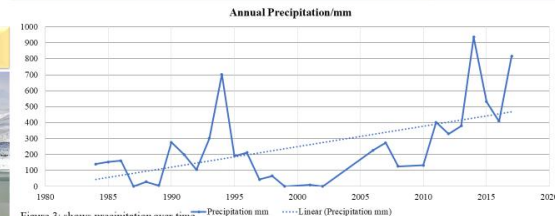


Figure 3: shows precipitation over time

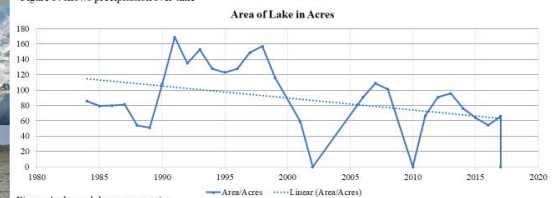


Figure 4: shows lake area over time

The long-term focus of this project is the realization of the temporal changes in Lake and to explore possible reasons of its shrinkage.

Lake area vs. precipitation

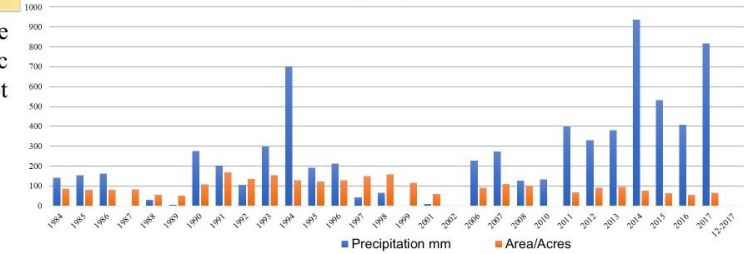


Figure 5: shows precipitation vs Lake surface area over time

- Figure 4 shows the dynamic changes in the surface area of the Hanna Lake during 1984 – 2017.
- Hanna lake has completely dried thrice during 2002, 2010, and 2017.
- The total surface area of lake has decreased, whereas, the annual precipitation has increased during study period (Figure 3). Pakistan Metrological Department (PMD) also confirmed increase in the precipitation amount [4].
- Therefore, the drying of the Lake is apparently not influenced by precipitation and there are factors responsible for that.
- The possible reasons may be an overexploitation of Lake water for domestic uses and lowering of groundwater in the surrounding area.

CONCLUSION

- Integrated watershed management approaches should be applied to control the degradation of Hanna Lake.
- A policy is needed for water sharing among users that ensures water availability in the Hanna Lake in all seasons.

FUTURE WORK

Currently there are limited data for more extensive study and analysis of the factors affecting Lake water availability. Researchers should focus on temporal weather data, surface lithology, groundwater depletion, and hydro-geological aspects to identify the influencing factors.

REFERENCES

1. Dunya News, 'Quetta's Hanna Lake dries up', January 23, 2018, <http://dunya-news.tv/en/Pakistan/424677-Quettas-Hanna-Lake-dries-up>
2. Monitoring variations of inland lakes in the arid region of Central Asia Jie BAI, Xi CHEN, Liao YANG, Hui FANG
3. Baz Gul, a resident of Hanna valley *Tourists appeal for revival of Hanna Lake* By APP Published: February 11, 2018
4. Technical report no.PMD-22/2009 published in 2009.



ECONOMIC ASSESMENT OF DAMAGED MANGO FARMS DUE TO 2011 FLOOD IN MIRPUR KHAS DISTRICT, SINDH

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Mehran University

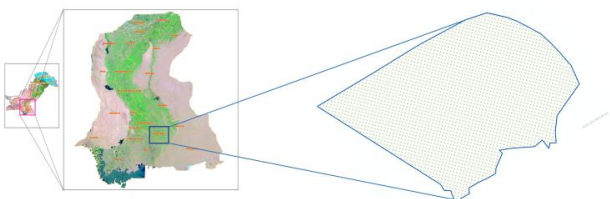
INTRODUCTION

- Mirpurkhas district of Sindh is famous for its delicious mango fruits throughout Pakistan with more than 200 varieties of mangoes.
- The Mirpurkhas Agriculture Department deputy director, Yar Mohammad Khaskheli said mango trees were spread over 33,000 acres in the district before 2011 flood and rains that destroyed almost 20 per cent of the total farms.
- In Mirpur Khas, most of the agricultural area was badly affected due to heavy rain, breach in the Mithrao Canal, and over flowing of the Left Bank Outfall Drain (LBOD) carrying saline water from different small channels.
- Flood water, that inundated around 1836.3 km² of the district area, not only damaged the mango farms, but affected people who were engaged in the mango business. Many lost their jobs and livelihoods.
- As a result, huge losses occurred to the national economy owing to low mango production in the following years.
- Poor land drainage and shallow water table also resulted in water logging and salinity that destroyed the already affected mango orchards.

OBJECTIVES

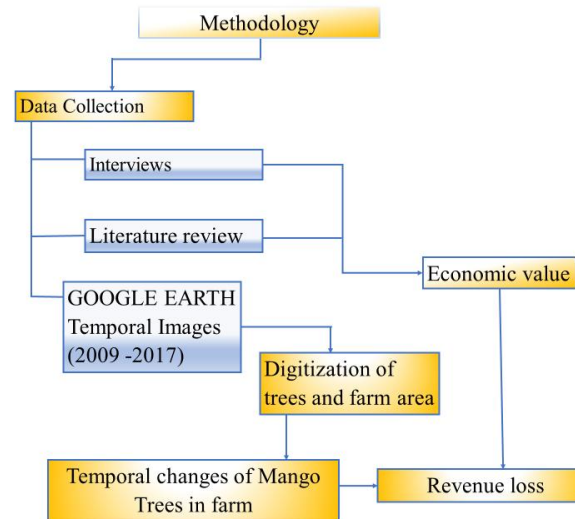
- To estimate area reduction in mango orchards due to 2011 flood using Google Earth historic images.
- To estimate economic damages of mango farms in the Mirpur Khas district due to 2011 flood.

STUDY AREA

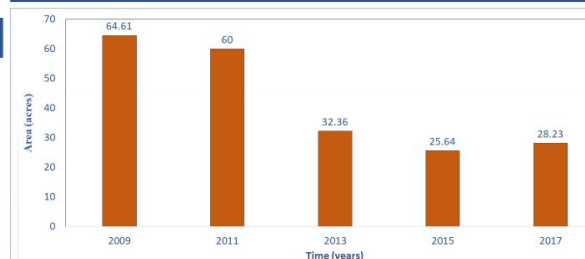


METHODOLOGY

Research data were collected from field, news papers, and Google Earth for damage assessment. Farm area and mango trees were digitized on map. Economic value was calculated using number of trees multiplied by the total annual income per tree. Economic loss was also calculated as the revenue lost in the flood damaged farm area.



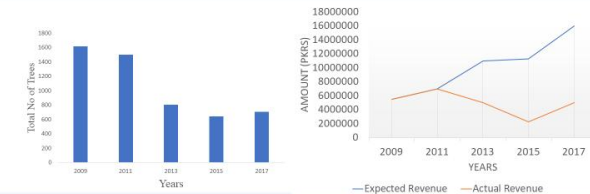
AREA COVERED WITH TREE (2009-2017)



GOOGLE EARTH IMAGES



RESULTS



CONCLUSION AND RECCOMEDATION

- Area covered with trees decreased due to flood.
- Retained water increased salinity and water logging, which is not favorable for mango tree.
- There is about 26 million PKRs total economic loss to the owner of land as flood impacted land.
- Land drainage is necessary to cope with such disaster.
- Government should take advance actions and should provide facilities to prevent national economic loss.

REFERENCES

- <http://lhdpmgma.blogspot.com/2011/12/sindh-floods-lbod-was-mistake-experts.html>
- Mateul, Memon Akhtar, Sher Muhammad, Siddiqi Paras, and Jillani Rahmatullah. 2012. "Techniques of Remote Sensing and GIS for Flood Monitoring and Damage Assessment: A Case Study of Sindh Province, Pakistan." *Egyptian Journal of Remote Sensing and Space Science* 15(2):135-41
- Allah Bux Khokhar (Farm Manager)

Course Instructor: Dr. Arjumand Zaidi
Hazard Planning and Risk Management, Spring 2018
Integrated Water Resource Management



ABDUL TAUQEER
FALAK NAZ
MAAZ SALEEM
ZAIN-UL-ABDIN

IMPACT ASSESSMENT OF LEFT BANK OFFFALL DRAIN A CASE STUDY OF NAWABSHAH

US-Pakistan Center for Advanced Studies in Water, MUET, Jamshoro.

SUBJECT: HAZARD PLANNING AND
RISK MANAGEMENT
SUPERVISOR: DR. ARJUMAND ZAIDI



INTRODUCTION

Water logging and salinity are main problems in Sindh. Rise in water table was due to poor drainage facilities when canal irrigation system of Indus plains was developed. Highly flat topography, the inadequate natural drainage, the semi-arid climate with high evaporation became cause of water logging (Steenbergen & Oliemans, 2002) and about 70 % of ground water of Sindh province is saline. In 1985 Government of Pakistan initiated a project Left Bank Outfall Drain (LBOD) in supervision of Water and Power Development Authority (WAPDA). LBOD-1 is the main project that provides drainage to districts Sanghar, Nawabshah, Mirpurkhas and Badin.

SOFTWARE	SATELLITE	YEARS
ERDAS, Arc GIS	LandSat 4-5	Feb, 1991
Arc GIS	LandSat 4-5	Feb, 2000
	LandSat 7	Feb, 2010
	LandSat 8	Feb, 2014
	LandSat 8	Feb, 2017

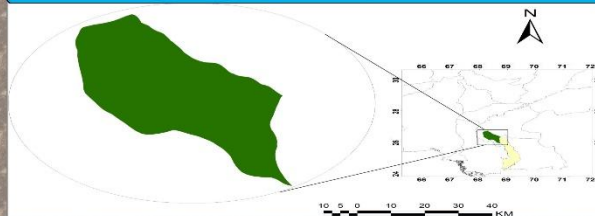
OBJECTIVE

To assess the impact of LBOD project in district Nawabshah by monitoring the change in vegetation for years 1991, 2000, 2010, 2014 and 2017.

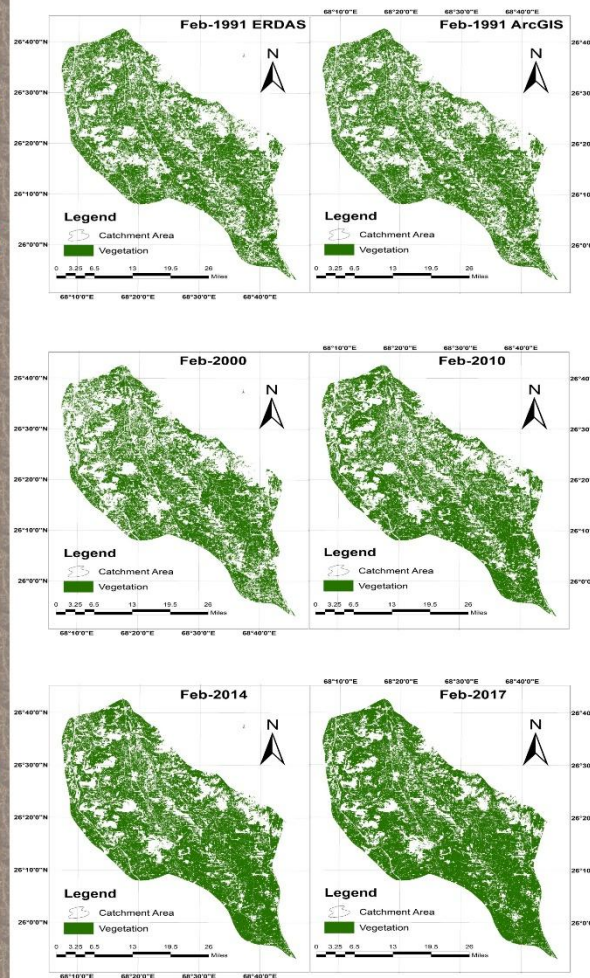
METHODOLOGY



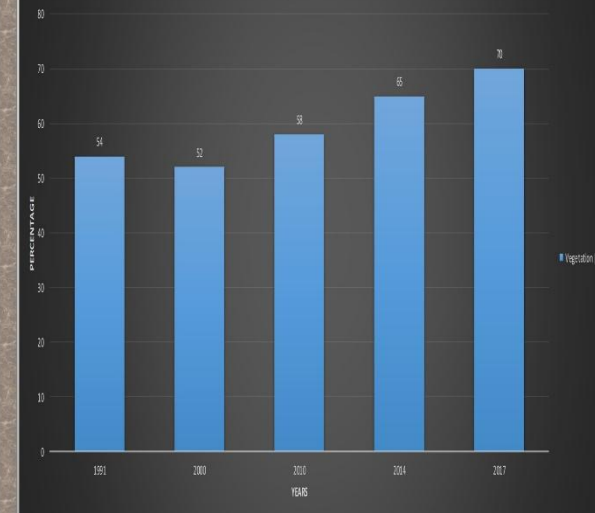
STUDY AREA



RESULTS



Vegetation Trend



CONCLUSION

The study shows the disaster mitigation project initiated by the WAPDA i.e. LBOD has remained successful as vegetation has increased from year 1991 to year 2017 for Nawabshah. Anomaly in year 2000 may possibly be due to the severe drought which hit Pakistan in 1999.

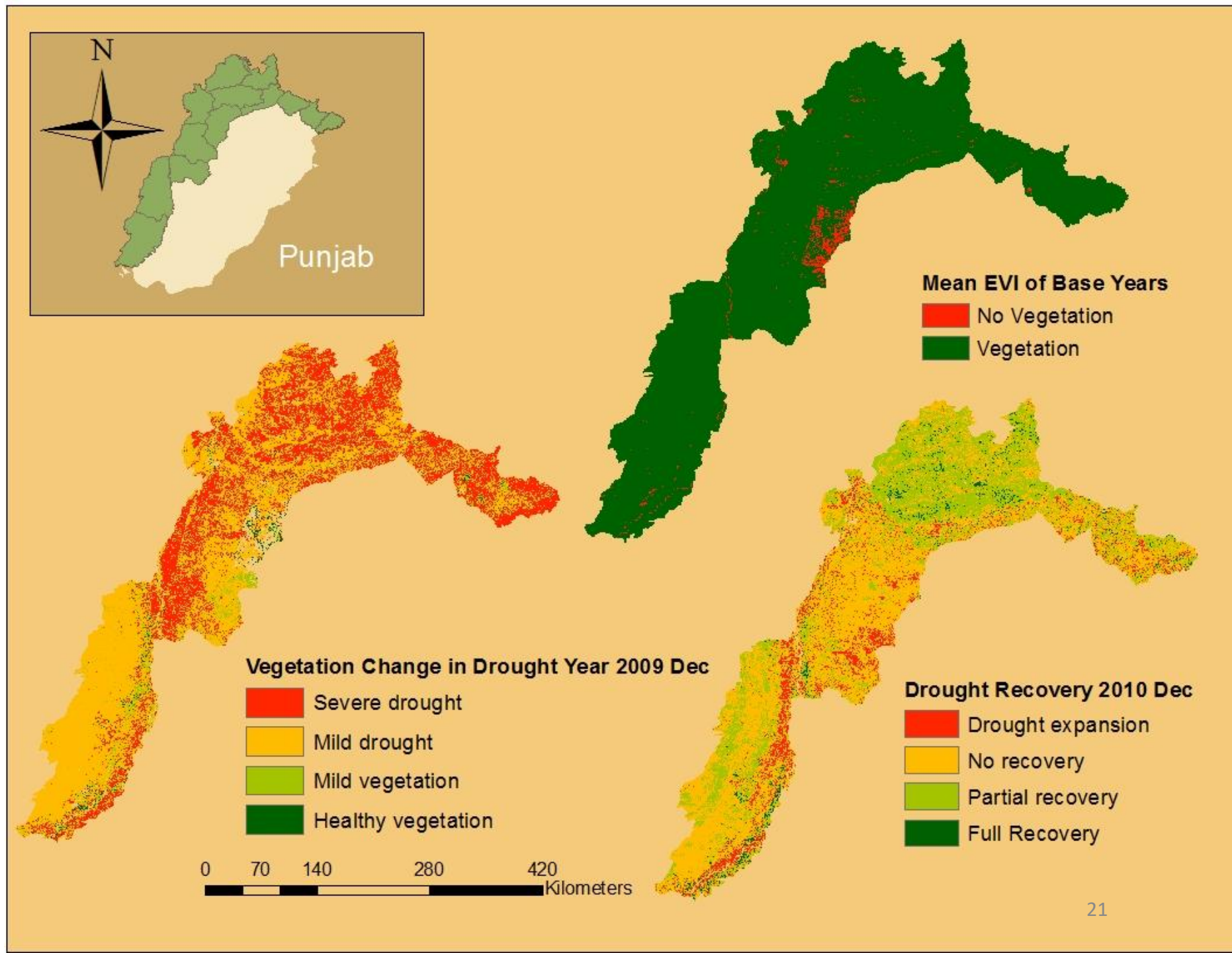
REFERENCES

Steenbergen, F. Van, & Oliemans, W. (2002). A review of policies in groundwater management in Pakistan 1950 – 2000. Water, 4, 323–V344.

ACKNOWLEDGEMENT

This is to acknowledge the efforts of Mr. Saad ul Haq, Mr. Rao Zahid Khalil, Mr. Nabeel Qureshi, Mr. Shoaib Jamro, Mr. Zaki Zaidi, Ms. Vengus Panhwar, Mr. Umair Bhatti, and Ms. Saima awan for helping in our class project.

DROUGHT IMPACT AND RECOVERY A CASE STUDY OF RAINFED AREA OF PUNJAB, PAKISTAN



2010

FLOODED ROADS



INTRODUCTION

The objective of this study is to identify the extent of road network affected during 2010 Floods in Sindh, Pakistan using geographic information system (GIS) software ArcGIS 10x.

The 2010 floods in Pakistan started in late July, resulting from heavy monsoon downpours over most parts of the country. The flood water inundated a large area of land in all provinces affecting 20 million people. This flood destroyed crops and submerged houses and infrastructure including road networks.

DISASTER MANAGEMENT

Flood hazard mapping is a constituent of disaster management (DM) cycle that can be utilized in all its phases.

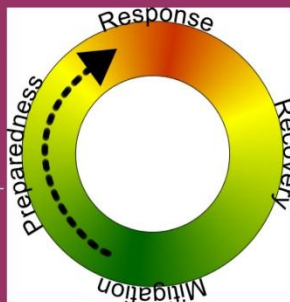
Mitigation: Hazard mapping is one of the basic needs for mitigation of any hazard. These maps will identify the flood hazard.

Preparedness: The extent of flood will demarcate the safe and vulnerable areas. This information will be helpful for evacuation planning and to well prepare for future flood disasters.

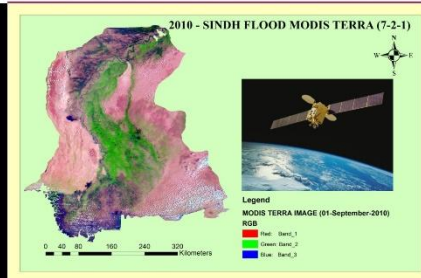
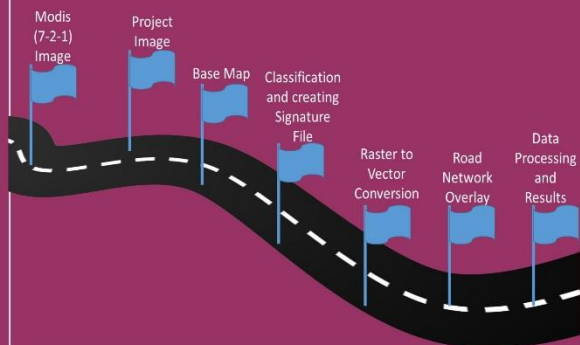
Response: Rapid flood mapping can lead to an efficient and focused response activities during flood disaster.

Recovery: The extent of damages can be identified for recovery purposes.

This mapping later serves as a decision support system for experts to improve the pitfalls of the system and apply proper mitigation measures.

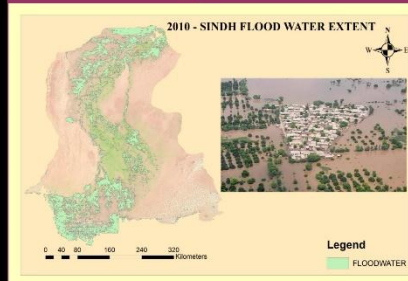
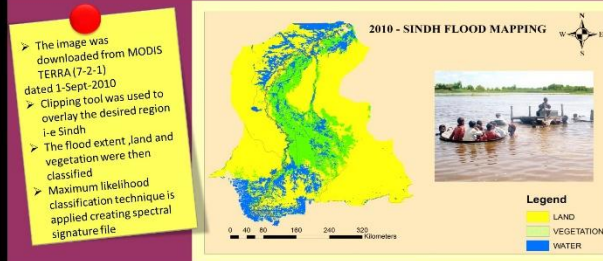


METHODOLOGY

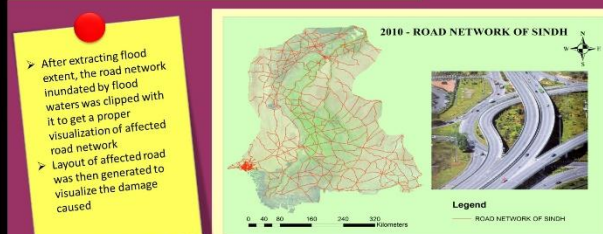


WHY ARCGIS?

- It takes Flood Management to an advanced level with key capabilities to manage the impacts of floods. We can:
- Generate authoritative flood extents
- Identify and analyze key infrastructure affected by flooding
- Present it clearly to aid decision makers to make better judgement



- After classification, raster data was converted to vector data to calculate flood area
- Flood extent was extracted using GIS tool

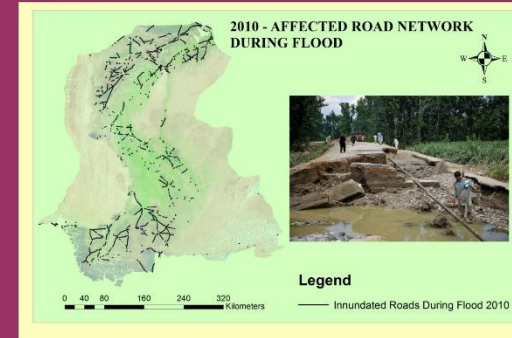


SUMMARY OF STEPS

1. Modis Image was obtained from NASA's website.
2. The image was clipped with Sindh boundary shapefile.
3. The clipped image was projected to WGS 84 UTM zone 42N.
4. Supervised image classification was carried out collecting training samples and developing spectral signatures for floodwater, land and vegetation.
5. The classified raster image was converted to vector data layer.
6. The road network was clipped with inundated flood to delineate affected roads during 2010 floods and a layout was developed.
7. This data can be useful for risk assessment for future floods.

CONCLUSION

The map of the affected road network will help identifying the vulnerable areas. The estimated total length of the affected road segments is **2,685 km**. These segments can hamper the relief activities by disrupting the link to the flood affected areas, and therefore, should be given priority while managing flood disaster.



PRESENTED AND DESIGNED BY:
FARIS AHMED BHATTI (16-MS-IWRM-08)
NABEEL ALI KHAN (16-MS-IWRM-03)

COURSE NAME: HAZARD PLANNING AND RISK MANAGEMENT
COURSE INSTRUCTOR: DR. ARJUMAND ZAIDI

DEPARTMENT: IWRM, USPCAS-W, MUET, Jamshoro

TAHIR ALI SHAIKH, MEHRAN SATTAR, VIPIN KUMAR OAD
U.S PAKISTAN CENTER FOR ADVANCED STUDIES IN WATER
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO

INTRODUCTION

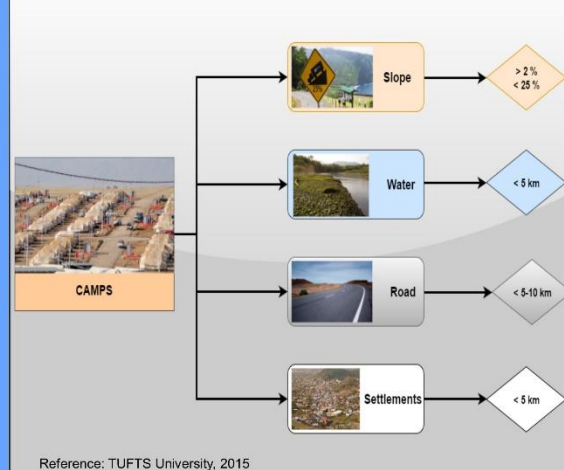
Pakistan Army launched a military operation, named Zarb e Azb, against various militant groups in North Waziristan Agency (NWA) on June 15, 2014. NWA is part of the Federally Administered Tribal Areas (FATA) which shares its west border with Afghanistan. The relocation of internally displaced people (IDPs) from the troublesome area began in early June 2014 to safe areas. The camp area at Baka Khel Bannu is among these IDPs relocation sites. As of March 16, 2015, a total of 1,024,24 families have been verified and registered as IDPs (NWA IDPs return intension survey). Out of which 57,143 families are residing in Baka Khel IDPs camps, Bannu. (Dawn 2014).

OBJECTIVES

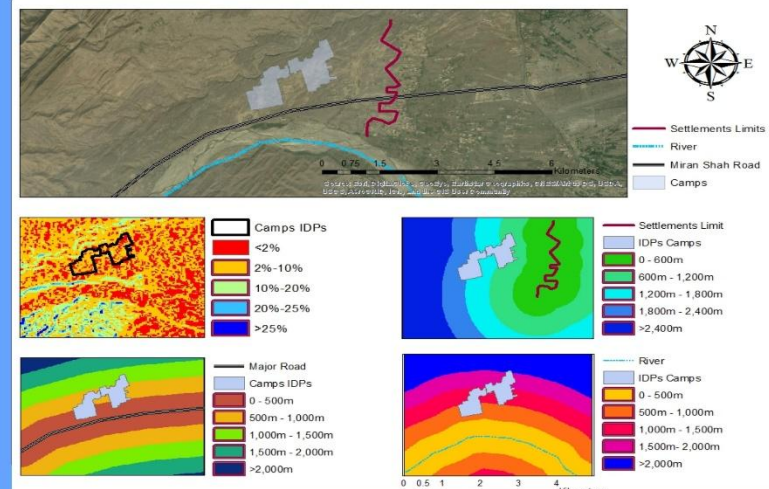
The objectives of this study are to (1) study satellite data derived indicators for selecting safe and suitable sites for IDPs, (2) to define criteria based on the threshold values of these indicators and (3) to assess against these criteria, the suitability of Baka Khel camp site.



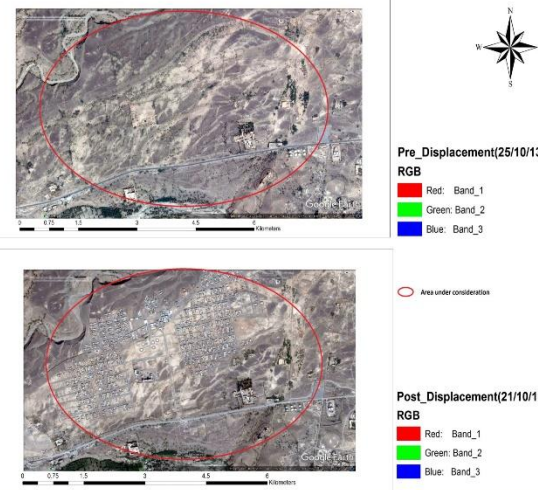
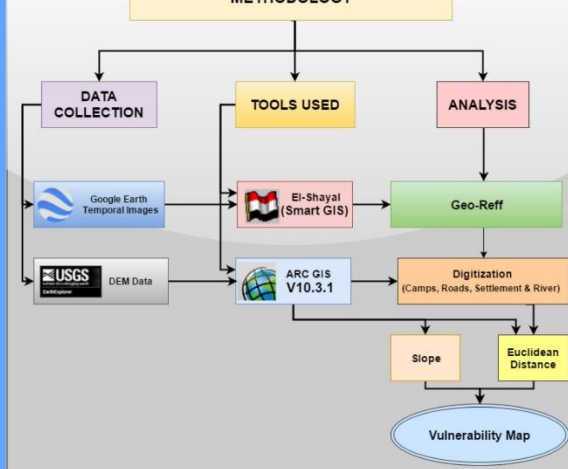
Criteria for Suitable Site



RESULTS



METHODOLOGY



DISCUSSIONS

The analysis was based on suitable surface slope value, and easy distances from major roads, water source (which is a nearby river in this case) and the nearest settlements/town. Although, some additional indicators could be included here, but due to unavailability of site specific data, only the satellite derived indicators were used. Generally people prefer to live in the closer vicinity of year round water supply, say around 5 km, so that they may reach there with ease. To provide them an easy access to a major road, a distance of 5 to 10 km is generally considered acceptable. Some researchers also talked about the reluctance of IDPs to live near people of different religion or ethnicity background. A fairly realistic assumption of IDPs having no such differences with the Bannu settlements may change this reluctance to a willingness to live near the settlements to fulfill their everyday needs. Ideally, the camp locations should have a slightly inclined ground surface for proper drainage but not as steep to make living difficult (Tufts, 2015).

CONCLUSIONS AND RECOMMENDATIONS

- The camps are in a closer vicinity to Miran Shah road (500 m to 1.5 km), which is a major road that connects Baka Khel union council to other nearby areas.
- The Baka Khel union council building is approximately 800 - 900 m away from the camp area which is not more than a walking distance.
- The source of water is also near the camps i.e. around 500 m to 2 km.
- The slope of the camp area is between 0 - 10 %. The unsuitable range of less than 2 % slope was found in the area but accepted since no other suitable slope area is available in the surrounding.
- It is concluded, therefore, that over all the camp site at Baka Khel is a suitable location.
- More indicators including socio-economic factors should also be investigated and included in future studies.
- Although the resources like water, settlements and road are not far away from the camp but still most of the people are willing to go back to their homes. The reasons for people willingness to leave the camp might be lack of food and water aid or their own areas are secure now.

REFERENCES

- Return intention survey NWA 2015.
- Dawn news paper report (2014).
- TUFTS University, 2015, Site Selection for Refugee and IDP Camps to House Population Displaced from Nigeria. https://sites.tufts.edu/gis/files/2015/05/Spink_Lauren_DHP207_2015.pdf



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TSUNAMI HAZARD MAPPING FOR GWADAR PORT USING GIS

US-Pakistan Center for Advanced Studies in Water, MUET, Jamshoro.

SUBJECT: HAZARD PLANNING AND RISK MANAGEMENT
INSTRUCTOR: DR. ARJUMAND ZAIDI



INTRODUCTION

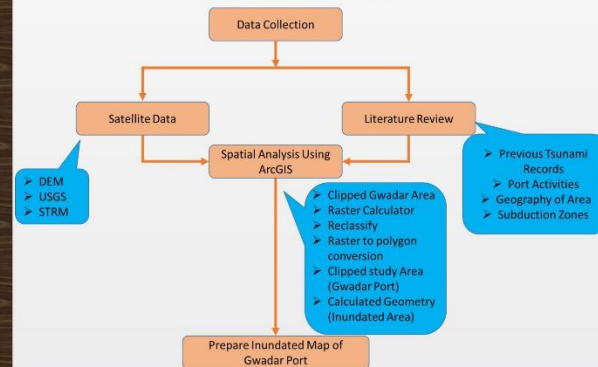
Gwadar port is the world's largest deep seaport constructed along the coastline of Baluchistan, Pakistan. Gwadar city is susceptible to tsunami hazard due to earthquakes generated at Makran Subduction Zone (MSZ) located offshore on the Arabian Sea. Tsunami usually generates in shallow waters less than 30 km deep due to large earthquakes in subduction zones. The MSZ exists between the Arabian and the Eurasian plates that have had many earthquake-generated tsunamis in the past. Any future large earthquake along MSZ has a potential of tsunami hazard for the entire coast of Pakistan including Gwadar city. To manage tsunami hazard and secure port activities, it is important to develop tsunami hazard maps of Gwadar port. This study is an attempt to map tsunami hazard in the port area using geographical information system software ArcGIS 10x. Different tsunami wave heights (Table 1) estimated through various simulation models by Pakistan Meteorological Department (PMD) are used for calculating the extent of inundation in the port area.

Table 1. Magnitude of earthquakes and Tsunami wave heights using different models

Software	Earthquake Magnitude	Tsunami Wave Height
Nakamura	Mw 8.5	8.8 m
MOST (Method Of Splitting Tsunami)	Mw 9.0	7.5 m
	Mw 8.5	4.2 m
	Mw 8.1	1.7 m
	Mw 7.7	0.5 m
GUIAR	Mw 9.0	5.4 m

Source: Pakistan Meteorological Department Karachi (PMD)

METHODOLOGY



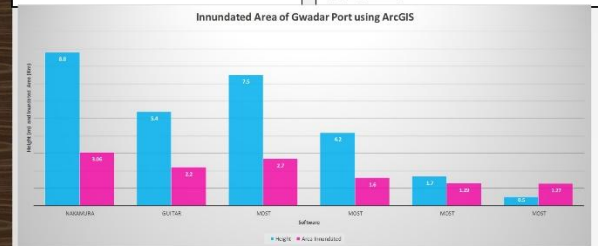
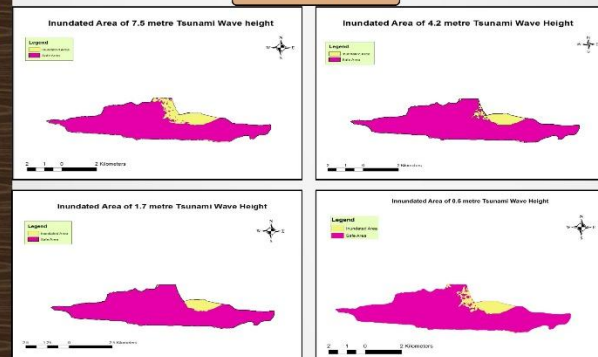
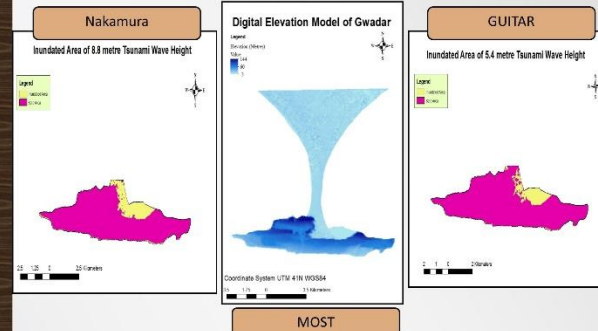
STUDY AREA



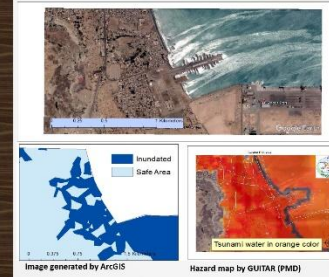
Gwadar is an important city of Makran Coast with a tsunami hazard. Its port has a great importance in the country due to its special economic zone (Mahmood, Khan, & Rafi, 2012).

RESULTS

Tsunami hazard maps show the predicted inundated and safe areas of Gwadar port using GIS tools, different wave heights, and satellite derived digital elevation model (DEM). PMD has run three (03) models to simulate different wave heights given the magnitude of the earthquakes. The names of these simulation models are Guitar, Nakamura, and MOST (method of splitting tsunami). The derived inundation areas for all given wave heights are presented in the following maps.



COMPARISON WITH GUIAR SIMULATED HAZARD MAP



The tsunami simulation models not only calculate the wave height for a given earthquake magnitude, but they also simulate inundation extent. The GUIAR model based hazard map for Gwadar port was compared with the ArcGIS developed map. These two methods estimated the water extent for 9.0MW earthquake and 5.4 m wave height. Both maps show the major area of the Gwadar port inundated, although the GUIAR map has a larger extent than GIS map. The GIS map also show some safe spots within the flooded area. A further detailed analysis may reveal the accuracy of the model results.

CONCLUSION

We evaluated the earthquake generated tsunami hazard for Gwadar port. We used ArcGIS 10x to estimate inundation area of Gwadar port due to different tsunami wave height scenarios. The study results show the inundation of major port area due to a future tsunami that has a potential of being triggered by an earthquake at MSZ. These maps identify the possibility of catastrophic damages to life and infrastructures at the Gwadar port if a tsunami comes. The likelihood of inundation of the Gwadar port to its full extent is alarming, and therefore, proper mitigation and preparedness measures are needed to manage the risk of tsunami hazard at the port area in order to minimize the losses both in terms of property damages and due to disruption in the port activities.

RECOMMENDATIONS



REFERENCES

- Heidarzadeh, M., Pirooz, M. D., Zaker, N. H., Yalciner, A. C., Mokhtari, M., & Esmaili, A. (2008). Historical tsunami in the Makran Subduction Zone off the southern coasts of Iran and Pakistan and results of numerical modeling. *Ocean Engineering*, 35(8-9), 774-786. <https://doi.org/10.1016/j.oceaneng.2008.01.017>
- Mahmood, N., Khan, K., & Rafi, Z. (2012). Mapping of Tsunami Hazard along Makran Coast of Pakistan.

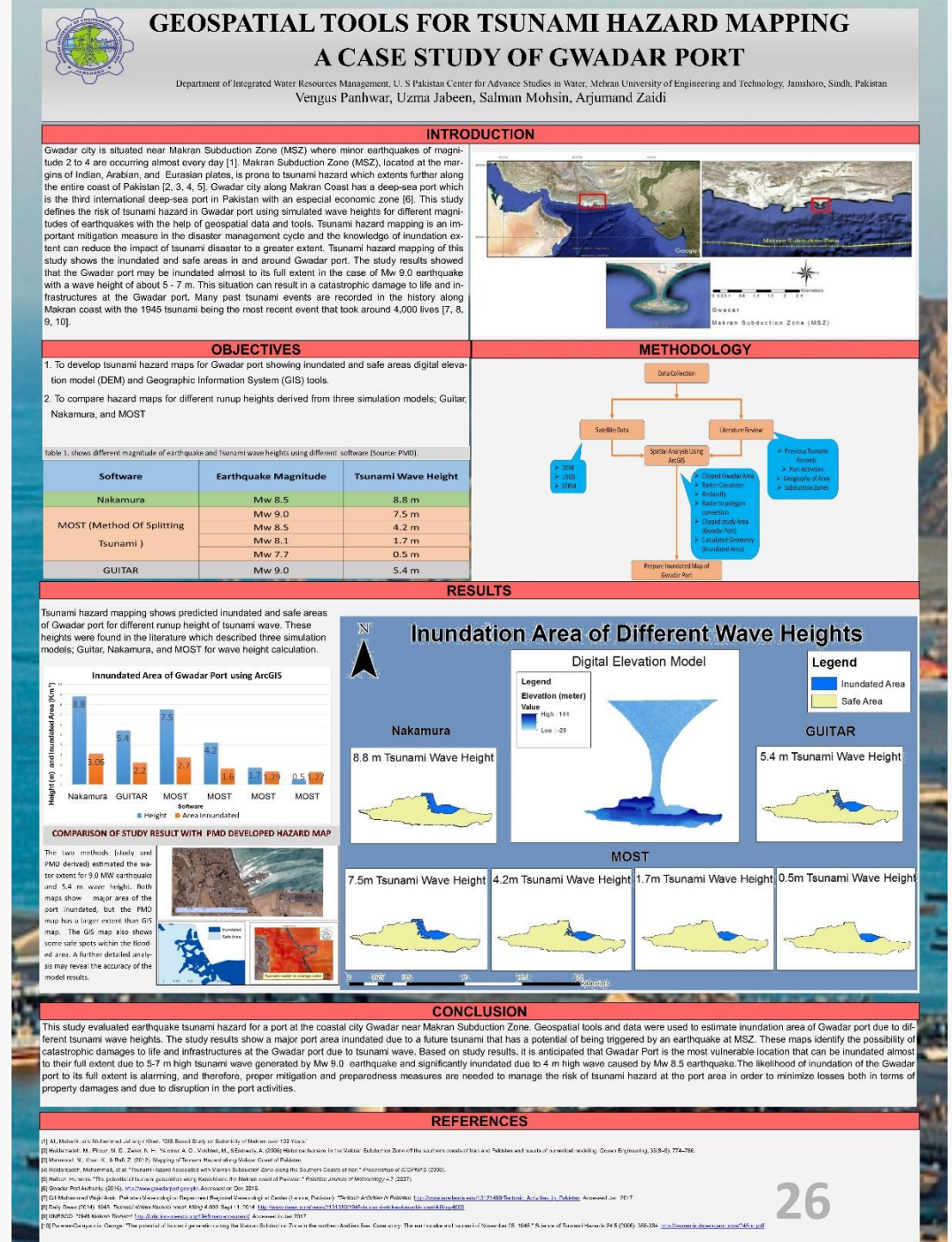
STUDENTS ACHIEVEMENTS

- Flooded Roads: Accepted for oral presentation
- Tsunami Hazard Mapping: Poster acceptance



GeoMundus 2017 Institute for Geoinformatics (IFGI), University of Münster, Germany 10th and 11th November

Ms. Vengus Panhwar
was one of the 3 winners of the
conference travel grant award

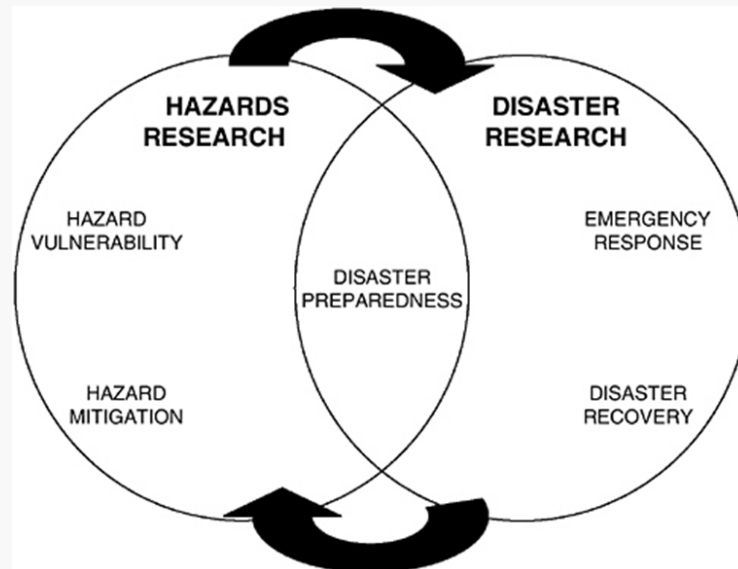


Terminologies, Definitions, Prevailing Theories, and Basic Concepts



HAZARD & DISASTER

What is the difference between them???



What is meant by a Hazard?

A dangerous condition or event, that threatens or has the potential for causing injury to life or damage to property/environment.



Definitions: Hazards

- A potentially damaging physical event, phenomenon, or human activity that may cause the loss of life or injury, property damage, social and economic disruption, or environmental degradation (UNISDR).
- The probability of occurrence within a specified period of time and within a given area of a potentially damaging phenomenon" (UNDRO, 1991).

Hazard Categories

- Two main categories
 - Natural
 - Caused because of natural processes or phenomena within earth system
 - Manmade
 - Due to human negligence

Some are Socio-natural

- Both natural and manmade
- Flood may be due to both excessive rain and change in land use practices

Examples

- Geological hazards
- Water & climatic
- Environmental
- Manmade or technological

- ▶ Examples
 - ▶ Earthquake
 - ▶ Volcanic eruption
 - ▶ Tsunami
 - ▶ Landslide
 - ▶ Floods
 - ▶ Droughts
 - ▶ Cyclone
 - ▶ Environmental pollution
 - ▶ deforestation
 - ▶ Forest fire
 - ▶ War
 - ▶ Industrial accidents
 - ▶ Oil spills
 - ▶ etc

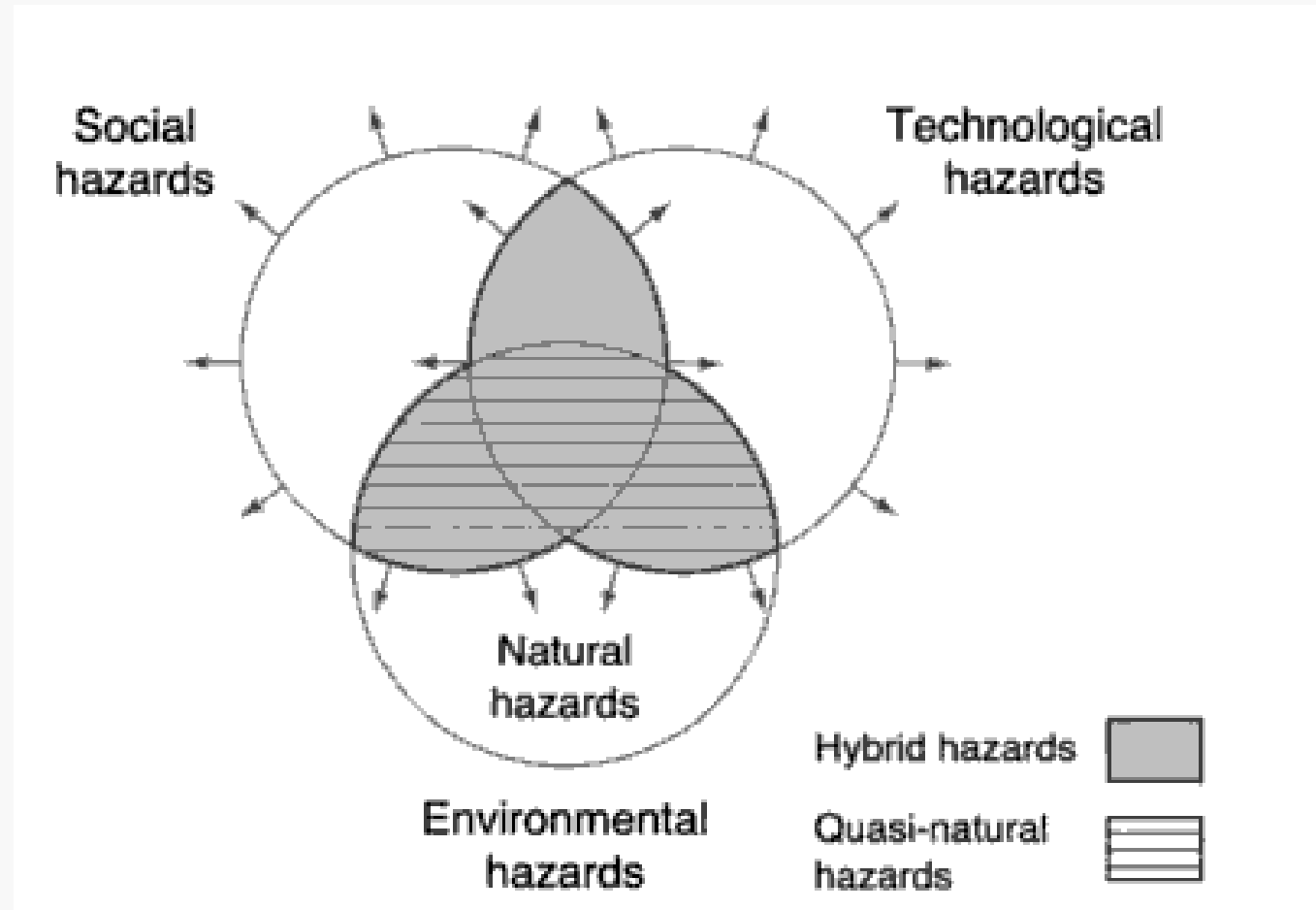
Note: These categories are not mutually exclusive

What do you think about Global warming?

Is this a hazard?

- If so, then what type of hazard - environmental or manmade or ... ?
- Global warming and temperature rise may cause many other hazards - like?
 - Glacier retreat, floods, droughts, forest fire, hurricanes, cyclones, etc.

Hazard Spectrum (Jones, 1993)



If human activity has caused the natural hazard then it can be classified as a **quasi-natural**: Examples???

Na-tech: natural **hazards** triggering technological Disasters: Examples??



Disaster?



Meaning of the word 'Disaster'

Meaning has changed through times

- In early usage it had reference to *unfavorable or negative effects, usually of a personal nature, resulting from a star or a planet* (E. L. Quarantelli)
- *"we make guilty of our disasters the sun, the moon and stars"* - Shakespeare, King Lear-Act I, Scene 2, 1605

- Then there came a time when the word *disaster* was applied to major physical disturbances such as
 - earthquakes and
 - floods
 - *or what traditionally known as the Acts of God (supernatural)*
- Later the term '*Natural*' substituted for the term '*Supernatural*'
- In recent decades notion of "*Acts of Men*" going along with Acts of God
- Some social scientists phrased disasters as the manifestations of the vulnerabilities of *social systems*

Disaster?

- Sudden adverse or unfortunate extreme event which causes great damage to human being as well as plants and animals
- Either
 - natural or manmade
 - from intense to diffuse
 - predictable to unpredictable
- Disaster occurs:
 - Rapidly
 - Instantaneously
 - Indiscriminately

Disaster?

- Severity varies and can be measured in absolute terms:
 - Deaths
 - Economic damages
 - Number of people affected
- Impact depends on capability of area to absorb the destructive effects.
- An event causing disaster at local level may not be impacting at regional level.

Disaster - Definition by the United Nations

A serious disruption of the functioning of a community or a society involving widespread human, material, economic or environmental losses and impacts, which **exceeds the ability of the affected community or society to cope using its own resources** (UNISDR)

Disaster

- A disaster occurs when a significant number of vulnerable people experience a hazard and suffer severe damage and/or disruption of their livelihood system in such a way that the recovery is unlikely without an external aid. (Blaikie, 1994)
- A situation or event, which overwhelms local capacity, necessitating a request to national or international level for external assistance; an unforeseen and often sudden event that causes great damage, destruction and human suffering (Centre for Research on the Epidemiology of Disasters)

Reading Assignment

Read

1. <https://www.bbc.com/news/world-asia-56252695>
2. <https://wellbeingintl.org/climate-change-and-cop26/>
3. <https://wellbeingintl.org/cop-28-running-out-of-time/>

Discussion/Comment/Question

